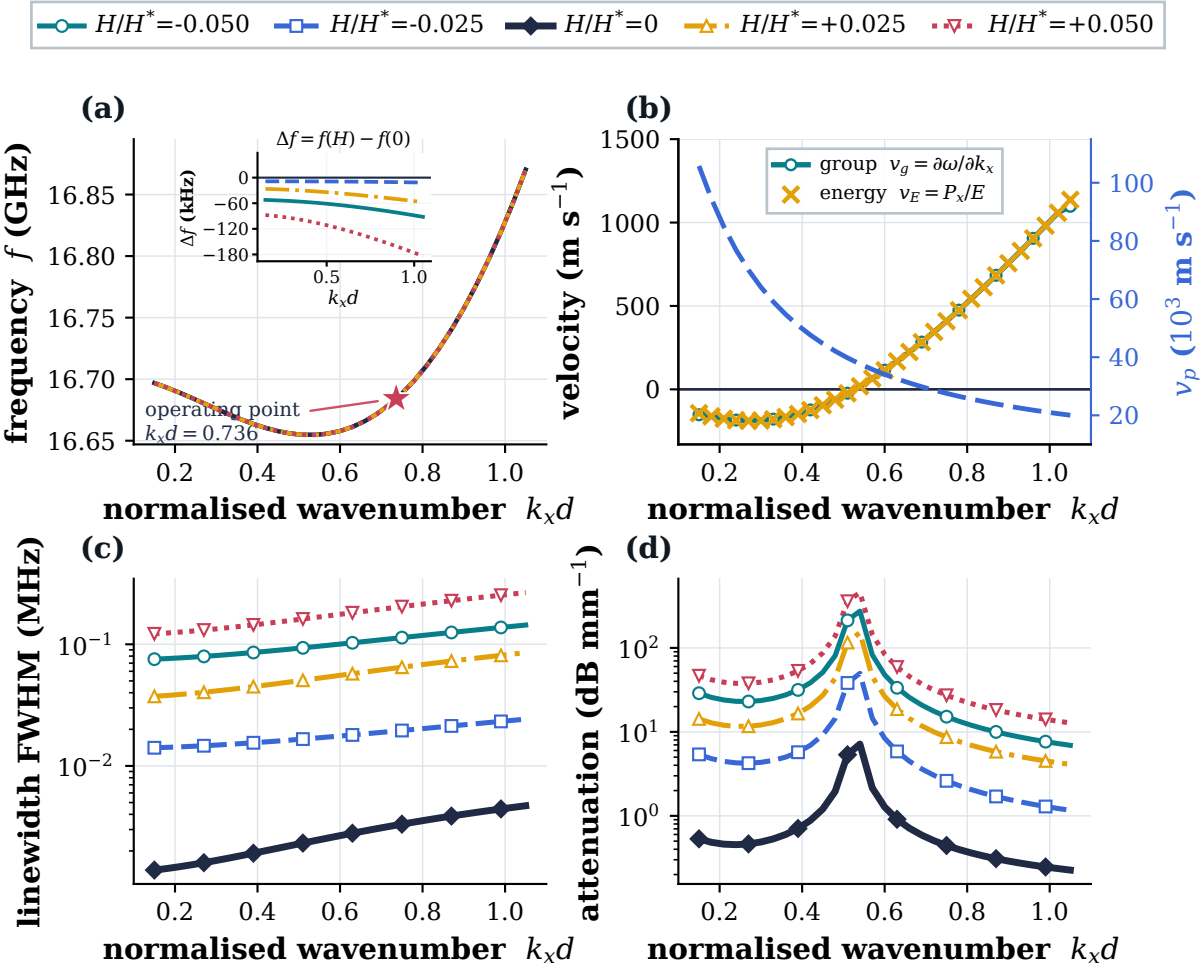
H/H^*=-0.050
H/H^*=-0.025
H/H^*=0
H/H^*=+0.025
H/H^*=+0.050
(a)
frequency f (GHz)
16.85
16.80
16.75
16.70
16.65
$\Delta f = f(H) - f(0)$
Δf (kHz)
0
−60
−120
−180
0.5
1.0
$k_x d$
operating point
$k_x d = 0.736$
0.2
0.4
0.6
0.8
1.0
normalised wavenumber $k_x d$
(b)
velocity (m s^{-1})
1500
1000
500
0
group $v_g = \partial\omega/\partial k_x$
energy $v_E = P_x/E$
v_p (10^3 m s^{-1})
100
80
60
40
20
0.2
0.4
0.6
0.8
1.0
normalised wavenumber $k_x d$
(c)
linewidth FWHM (MHz)
10^{-1}
10^{-2}
0.2
0.4
0.6
0.8
1.0
normalised wavenumber $k_x d$
(d)
attenuation (dB mm^{-1})
10^2
10^1
10^0
0.2
0.4
0.6
0.8
1.0
normalised wavenumber $k_x d$